

Omar Ibne Shahid

Lecturer, Department of Electrical and Computer Engineering
North South University, Dhaka, Bangladesh

Research interests: AI in healthcare, medical informatics, biomedical signal processing, explainable AI

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Research experience

- Analyzing Stroke Patient Impairment** May 2025 – Present
University of Alabama at Birmingham Supervisor: **Dr. Brandon Oubre**
- Translation of a Wearable Fetal Movement Monitor Towards Stillbirth Prevention** Mar 2023 – Dec 2023
Wellcome Leap In Utero, University College Dublin (remote) Supervisors: **Dr. Niamh Nowlan** and **Dr. Abhishek Kumar Ghosh**
- Machine Learning Based Physical Fatigue Detection System** Oct 2021 – Jan 2023
The State University of New York at Buffalo Supervisor: **Dr. Lora Cavuoto**
- Human Activity Recognition Using IMU Sensor Unit** Jan 2020 – Dec 2020
University of East London (remote) Supervisor: **Dr. Md. Atiqur Rahman Ahad**

Employment

- Lecturer** Jan 2024 – Present
Department of Electrical and Computer Engineering, North South University, Dhaka
Courses: Pattern Recognition and Neural Networks; Design and Analysis of Algorithms; Data Structures and Algorithms; Digital Logic Design.
- Research Assistant (remote)** Apr 2023 – Dec 2023
Wellcome Leap In Utero, University College Dublin
- Graduate Teaching Assistant** Aug 2021 – Dec 2022
Department of Industrial and Systems Engineering, University at Buffalo
- Engineer, Level-I** Feb 2020 – Jul 2021
DataSoft Manufacturing & Assembly Inc. Limited Embedded systems and artificial intelligence
- Intern Engineer** Jul 2019 – Oct 2019
DataSoft Manufacturing & Assembly Inc. Limited Machine learning and AI development
- Mentor, Bangladesh Robot Olympiad National Team** Jul 2018 – Dec 2019
Coordinated by Dr. Lafifa Jamal, Central Committee Member, International Robot Olympiad

Education

- MSc in Human Factors Engineering** Aug 2021 – Jan 2023
Department of Industrial and Systems Engineering, The State University of New York at Buffalo, USA
CGPA 3.50/4.00. Master's research: physical fatigue detection; UX design for older adults.
- BSc in Robotics and Mechatronics Engineering** Feb 2016 – Dec 2019
University of Dhaka, Bangladesh
CGPA 3.72/4.00. Undergraduate project: deep-model-based detection of autism spectrum disorder (ASD) through activity recognition.

Publications

- Journal article**
- Comparative Performance Evaluation of Fetal Movement-Detecting Wearable Sensors Using a Body-Worn Device** 2024
A. K. Ghosh, **O. I. Shahid**, N. C. Nowlan, R. Vaidyanathan
IEEE Sensors Journal, vol. 24, no. 17, pp. 28018–28027 [Link](#)
- Conference papers**
- Classifying Eye-Tracking Data Using Saliency Maps** 2021
S. Rahman, S. Rahman, **O. Shahid**, M. T. Abdullah, J. A. Shourov
25th International Conference on Pattern Recognition (ICPR), pp. 9288–9295 [Link](#)

UPIC: User and Position Independent Classical Approach for Locomotion and Transportation Modes Recognition 2020
M. S. Siraj, M. A. A. Faisal, **O. Shahid**, F. F. Abir, T. Hossain, S. Inoue, M. A. R. Ahad
Adjunct Proceedings of ACM UbiComp/ISWC 2020, pp. 340–345 [Link](#)

Vision-based Action Understanding for Assistive Healthcare: A Short Review 2019
M. A. R. Ahad, A. D. Antar, **O. Shahid**
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pp. 1–11 [Link](#)

Book chapters

Contactless Human Activity Analysis: An Overview of Different Modalities 2021
F. F. Abir, M. A. A. Faisal, **O. Shahid**, M. U. Ahmed
In Contactless Human Activity Analysis, Springer, pp. 83–112 [Link](#)

Cooking Activity Recognition with Varying Sampling Rates Using Deep Convolutional GRU Framework 2020
M. S. Siraj, **O. Shahid**, M. A. R. Ahad
In Human Activity Recognition Challenge, Springer, Singapore, pp. 115–126 [Link](#)

Under review

Single Sensor Estimation of Upper-Limb Stroke Impairment during Activities of Daily Living using Point-to-Point Submovement Kinematic
IEEE Journal of Biomedical and Health Informatics

SwinBact Net: A Transformer-Based Deep Framework for Bacterial Image Segmentation
Nature Scientific Reports

HumP-KD: A Hybrid Uncertainty-Aware Multi-Stage Progressive Knowledge Distillation Framework for Efficient Fire Classification
SN Computer Science

Geometric Deformation Invariance in Data-Efficient Point Cloud Classification Using Linear Optimal Transport Subspaces
Pattern Recognition Letters

Reviewer Experience

International Conference on Pattern Recognition (ICPR) 2026
ACM Conference on Computing and Sustainable Societies (COMPASS) 2026
Journal of Expert Systems with Applications 2026

Honors and awards

2nd Runner-up, Hardware Project Showcase Competition, University of Dhaka 2019
Top 3 Mentors, Bangladesh Robot Olympiad (Bangladesh Open Source Network) 2018
1st Runner-up, Business Idea Innovation Challenge, Bangladesh Innovation Forum 2017

Organizing and volunteering

Faculty Advisor Jan 2025 – Present
IEEE Robotics and Automation Society, North South University
Organizer 2018 and 2019
Bangladesh Robot Olympiad, national round
Treasurer 2018 – 2019
IEEE Robotics and Automation Society Student Branch, University of Dhaka Branch

References

Dr. Brandon Oubre
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Computer Science
University of Alabama at Birmingham
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Professor
Artificial Intelligence
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Dr. Niamh Nowlan
Professor
Biomedical Engineering
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